

# The Minimal Model Program

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# Outline of the talk

## 1 Introduction

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- 2 The MMP for 3-folds

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- $g \geq 2 \Rightarrow$  there is a  $3g - 3$ -parameter family of curves of general type.

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- One can use the above results to construct a moduli space.

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- Blow ups generate the birational equivalence relation for surfaces.
- Once again the geometry is understood in terms of the canonical line bundle  $\omega_X$ .

# Canonical line bundle

- If  $X$  is smooth of dimension  $n$ , then  $T_X$  denotes the **tangent bundle** of  $X$  and  $\omega_X = \Lambda^n T_X^\vee$  is the **canonical line bundle**.

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- $K_X$  denotes (a choice of) the **canonical divisor** i.e. a formal linear combination of codimension 1 subvarieties of  $X$  such that  $\omega_X = \mathcal{O}_X(K_X)$ .



# Birational invariants

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$$\kappa(X) := \text{tr.deg.}_{\mathbb{C}} R(X) - 1 \in \{-1, 0, 1, \dots, \dim(X)\}.$$

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- $\kappa(X) = 1 \Leftrightarrow$  there is a map  $X \rightarrow C$  whose general fiber is a curve of genus 1.
- If  $\kappa(X) = 2$ , then we have a good qualitative understanding of  $X$  (which we will now recall).

# Surfaces of general type

- By Castelnuovo's theorem we may repeatedly blow down  $-1$  curves  $E \subset X$  ( $\omega_X \cdot E = -1$ ,  $E \cong \mathbb{P}^1$ ) to get a sequence  $X \rightarrow X_1 \rightarrow X_2 \rightarrow \dots$



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- $\omega_{X_{min}}$  is semiample. In fact  $\phi_{\omega_{X_{min}}^{\otimes 5}} : X_{min} \rightarrow X_{can} \subset \mathbb{P}^N$  where  $X_{can} = Proj R(\omega_X)$  (note  $\deg(X_{can} \subset \mathbb{P}^N) = 25K_{X_{min}}^2$ ).

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- This allows us to construct moduli spaces for the minimal/canonical models of surfaces of general type and fixed  $K_{X_{\min}}^2$ .

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  - Threefolds
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*If  $\kappa(X) \geq 0$ , then there is a birational map  $X \dashrightarrow X_{\min}$  such that  $\omega_{X_{\min}}$  is nef (i.e.  $\omega_{X_{\min}} \cdot C \geq 0$  for any curve  $C \subset X_{\min}$ ).*

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- $X_{\min}$  is not unique and is (mildly) singular.

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- Flips are a new operation that we will discuss below.
- The minimal model  $X_{\min}$  is not unique.
- However, any two minimal models are related by a finite sequence of “elementary” operations known as flops.

# Running a MMP

- A **Minimal Model program** is a sequence of geometric operations  $X_i \dashrightarrow X_{i+1}$  that starting from a projective variety  $X$  produces a minimal model (if  $\kappa(X) \geq 0$ ) or a covering family of rational curves (if  $\kappa(X) < 0$ ).

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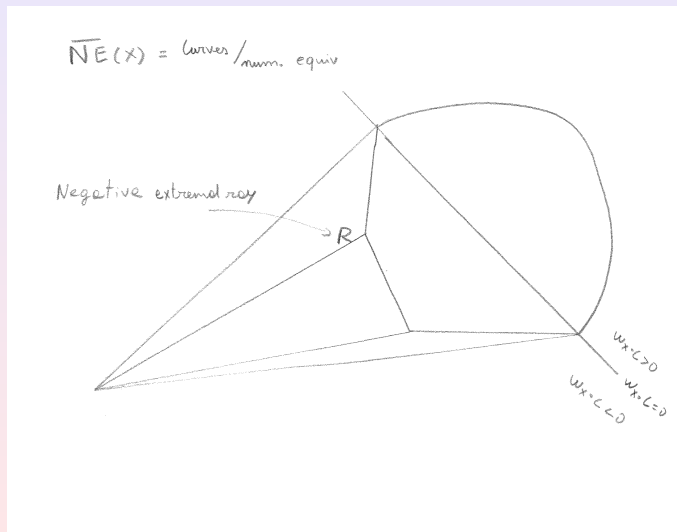
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- By the Cone Theorem, there is a negative extremal ray  $R = \mathbb{R}^+[C]$  where  $C \subset X$  is a curve such that  $\omega_X \cdot C < 0$  and there is a morphism  $\text{Cont}_R : X \rightarrow Z$  such that a curve  $D \subset X$  is contracted if and only if  $[D] \in R^+[C]$ .

# The Cone Theorem



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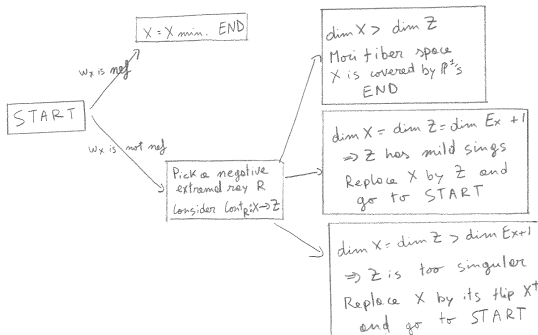
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- Instead of replacing  $X$  by  $Z$ , we will replace  $X$  by its **flip**  $X^+$ .



## MMP flow chart



# Flips

- The **flip** of  $X \rightarrow Z$  is a small birational morphism  $X^+ \rightarrow Z$  such that
  - 1)  $X \dashrightarrow X^+$  is an isomorphism on the complement of  $\text{Exc}(X \rightarrow Z)$ ,
  - 2)  $X^+$  has mild singularities, and
  - 3) If  $C \subset \text{Exc}(X^+ \rightarrow Z)$ , then  $\omega_{X^+} \cdot C > 0$ .

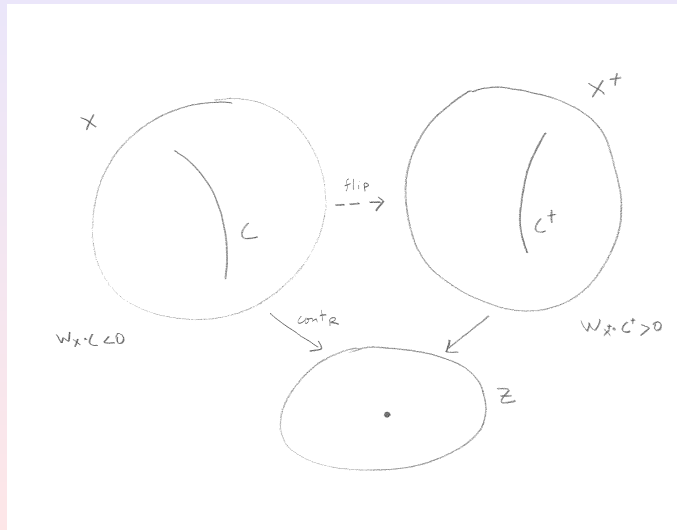
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- In other words, the flip  $X \dashrightarrow X^+$  is a surgery that replaces  $\omega_X$  negative curves by  $\omega_{X^+}$  positive curves.
- If the flip of  $X \rightarrow Z$  exists, it is unique and given by  $\text{Proj}_Z R(X)$ . So that to construct the flip we must show that  $R(X)$  is finitely generated over  $Z$  (i.e. if  $Z = \text{Spec}(A)$ ,  $R(X)$  is a finitely generated  $A$ -algebra).

# Picture of a flip



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- If  $X \rightarrow Z$  is a divisorial contraction, then  $\rho(Z) = \rho(X) - 1 \geq 0$ . If  $X \dashrightarrow X^+$  is a flip, then  $\rho(X^+) = \rho(X)$ . Therefore, it suffices to show that there is no infinite sequence of flips.



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- In dimension 3, this is relatively easy to do.

# Outline of the talk

- 1 Introduction
- 2 The MMP for 3-folds
- 3 Higher dimensional MMP

# Finite generation

Recently there has been substantial progress in higher dimensions:

Theorem (Birkar-Cascini-Hacon-M<sup>c</sup>Kernan and Siu)

*Let  $X$  be a smooth projective variety, then  $R(X)$  is finitely generated.*

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The two proofs are independent. Our proof is algebraic and uses the ideas of the MMP (minimal model program). Siu's proof is analytic and requires  $X$  to be of general type.

# Minimal models

Theorem (Birkar-Cascini-Hacon-M<sup>c</sup>Kernan)

*Minimal models  $X \dashrightarrow X_{\min}$  for smooth varieties of general type (i.e.  $\kappa(X) = \dim(X)$ ) always exist.*

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To prove this we show that:

- 1) Flips always exist;
- 2) Any appropriately chosen sequence of flips and divisorial contractions terminates.

# Parameter spaces

We can also generalize results necessary for the construction of moduli spaces of varieties of general type.

**Theorem (Tsuji, Hacon-M<sup>c</sup>Kernan, Takayama)**

*For any  $n > 0$  there exists an integer  $M = M(n)$  such that if  $X$  is a smooth variety of general type, then  $\phi_{\omega_X^{\otimes m}}$  is birational for all  $m \geq M$ .*



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Moreover, we can also show that:

## Theorem

*There is a parameter space for the set of  $n$ -dimensional varieties  $X$  with mild (canonical singularities) such that  $\omega_X$  is ample and  $K_X^n$  is fixed.*

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- (Birkar-Cascini-Hacon-M<sup>c</sup>Kernan) If flips exist in dimension  $n + 1$  (and minimal models exist in dimension  $n$ ) then minimal models exist in dimension  $n + 1$ . (Uses the ideas of the MMP and in particular techniques developed by Shokurov.)

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- Since the fibers have smaller dimension, it is reasonable to hope that finite generation will follow from the MMP in dimension  $n - 1$ .



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- By a compactness argument we show that there are only finitely many such min. models and hence the sequence of flips terminates.

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- Hacon-McKernan-Xu (work in progress) show the full ACC for LCTs for rational accumulation points.

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- The MMP in characteristic  $p > 0$ .